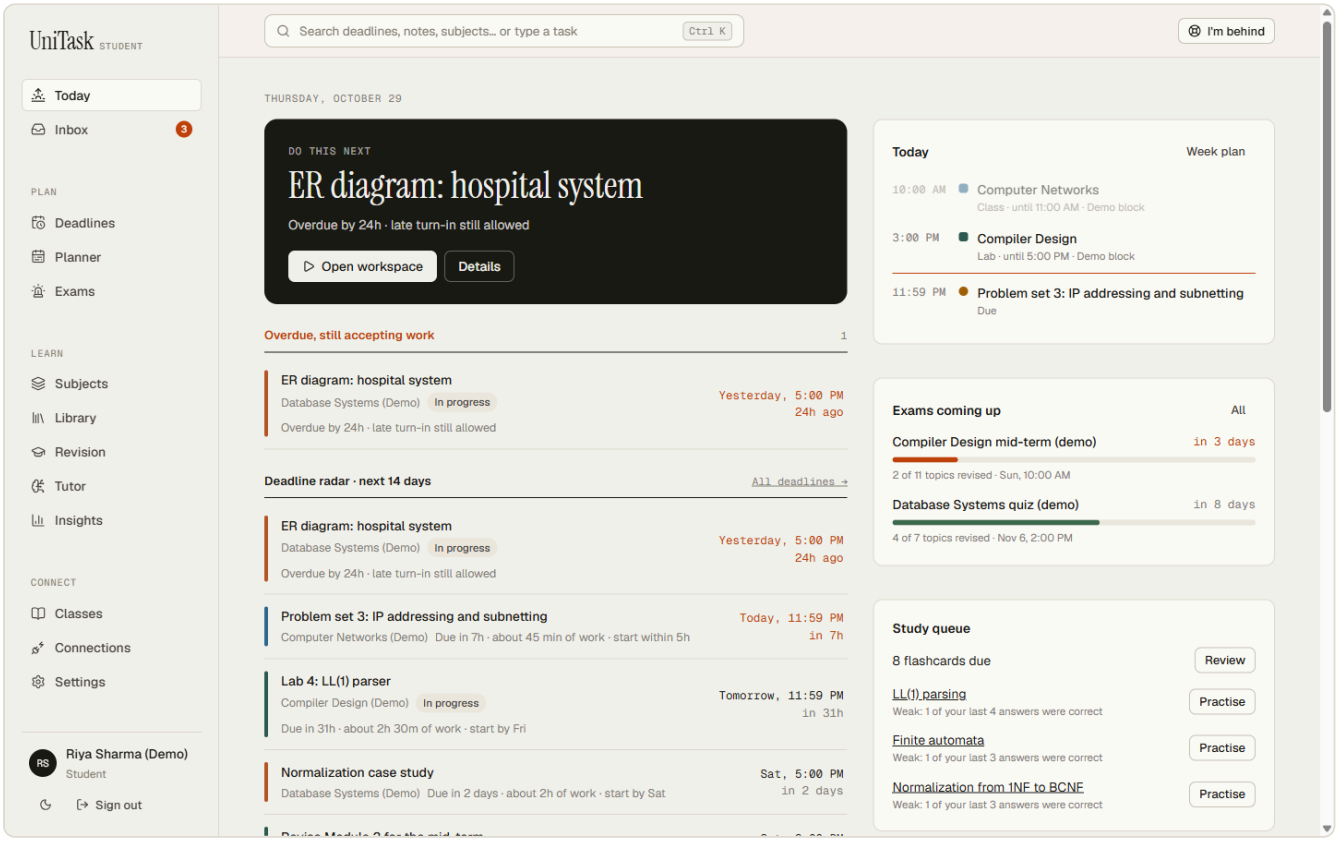
UniTask ACADEMIC ASSISTANT

Know what to do *next* — and actually learn it.

UniTask is a student's academic assistant. It pulls deadlines in automatically, decides what to do next, plans the week around real classes, rescues exam preparation when time is short, and teaches from the student's own notes with a tutor that adapts to how that student learns.



The Today screen: one clear next action, the day, exams and what to study.

Feature guide · every screen, what it does and how it helps the student · 29 October 2026 · screenshots from the running app with synthetic demo data

01 UniTask in one page

Students don't fail for lack of a to-do list. They miss deadlines that are scattered across Teams, Moodle and WhatsApp. They study the wrong topic the night before an exam. They lose the one page in their notes that explained the concept. And generic AI answers don't match their syllabus or their level. UniTask is built around those moments.

What a student gets

- **One next action**, with the reason, the effort and the time left.
- **Deadlines that arrive by themselves**: Teams, a Moodle calendar feed, a pasted notice or a file import.
- **A week plan** that fits work around classes and never schedules anything after its deadline.
- **Exam Rescue**: the best use of the hours left, using past papers and what's already covered.
- **A library that answers**: search inside notes and get answers that cite the page.
- **A personal tutor** that remembers misconceptions and checks understanding.
- **Revision on evidence**: flashcards and weak topics from real answers, not invented scores.

The principles

- **The student is the product.** Teacher tools exist but stay out of the student's way.
- **Decide, don't just display.** Every list is ranked and every ranking says why.
- **Never make things up.** Citations are checked against the actual pages; extracted dates are marked unconfirmed.
- **Calculated, not guessed.** Feasibility, schedules and priorities come from deterministic code. The AI explains and teaches.
- **Works without the cloud.** If an integration or the AI is unavailable, the app degrades honestly instead of breaking.

Contents

3. Sign in and demo entry
4. Today: the academic command centre
5. Command bar (Ctrl K)
6. “I'm behind”: recovery plan
7. Academic inbox
8. Deadlines
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16. Ask your notes, with page citations
17. Study desk
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19. Personal tutor
20. Insights
21. Connections: deadlines arrive by themselves
22. Settings: how you learn
23. Classes and the teacher side
24. On a phone

02 What is new here

The points to convey. Most student apps list deadlines or chat generically. UniTask decides, plans and teaches from the student's own material and record.

A tutor that knows this student

It adapts to the student's pace and preferred style and their own description of their background. It remembers misconceptions it has seen and checks understanding with a question. A wrong answer becomes a flashcard. The student can see and delete everything it remembers.

Answers that cite your pages

Every answer drawn from the student's notes links to the exact file and page. A citation the model invents is stripped before it reaches the student.

Exam Rescue

Given the hours left, it chooses topics by how often they appeared in the student's own past papers, how many other topics depend on them, and what's already covered. It says which topics don't fit and why, and never promises marks.

“I'm behind” recovery plan

One button turns panic into triage: what to do in the next three days, what to push, and where to ask for more time. It even drafts the extension email (never sent automatically).

Planner that respects reality

Free time is computed from the timetable and exams, with a daily cap. Work that can't fit before its deadline is listed, not silently dropped.

Past-paper intelligence

Questions are pulled out of uploaded papers, mapped to syllabus topics and grouped by repetition. History, not prediction, and the app says so.

Deadlines that find you

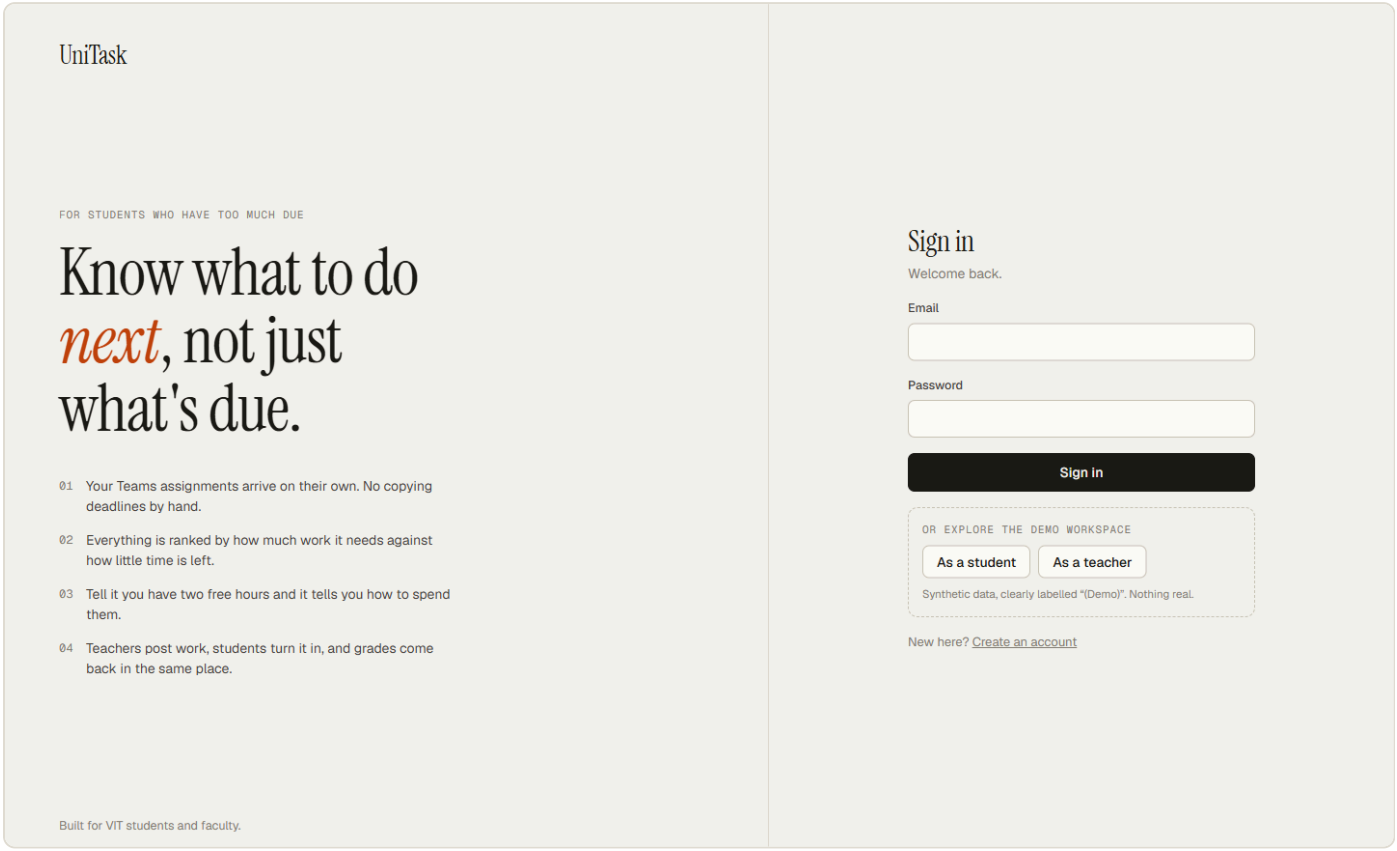
Teams assignments sync through Microsoft Graph. Moodle calendar feeds and files are ingested with de-duplication across platforms, and moved deadlines are highlighted.

Search and capture from anywhere

Ctrl K searches deadlines, subjects and the text inside every note. Typing “DBMS quiz Friday 5pm” captures it as a task with the date marked unconfirmed.

03 Sign in and demo entry

Students and teachers sign in to their own workspace. For presentations, the demo workspace opens in one click.



Sign-in, with one-click entry to the synthetic demo workspace

WHAT IT SHOWS

- Email sign-in and registration as student or teacher.
- Optional Amazon Cognito accounts with email verification.

HOW IT HELPS THE STUDENT

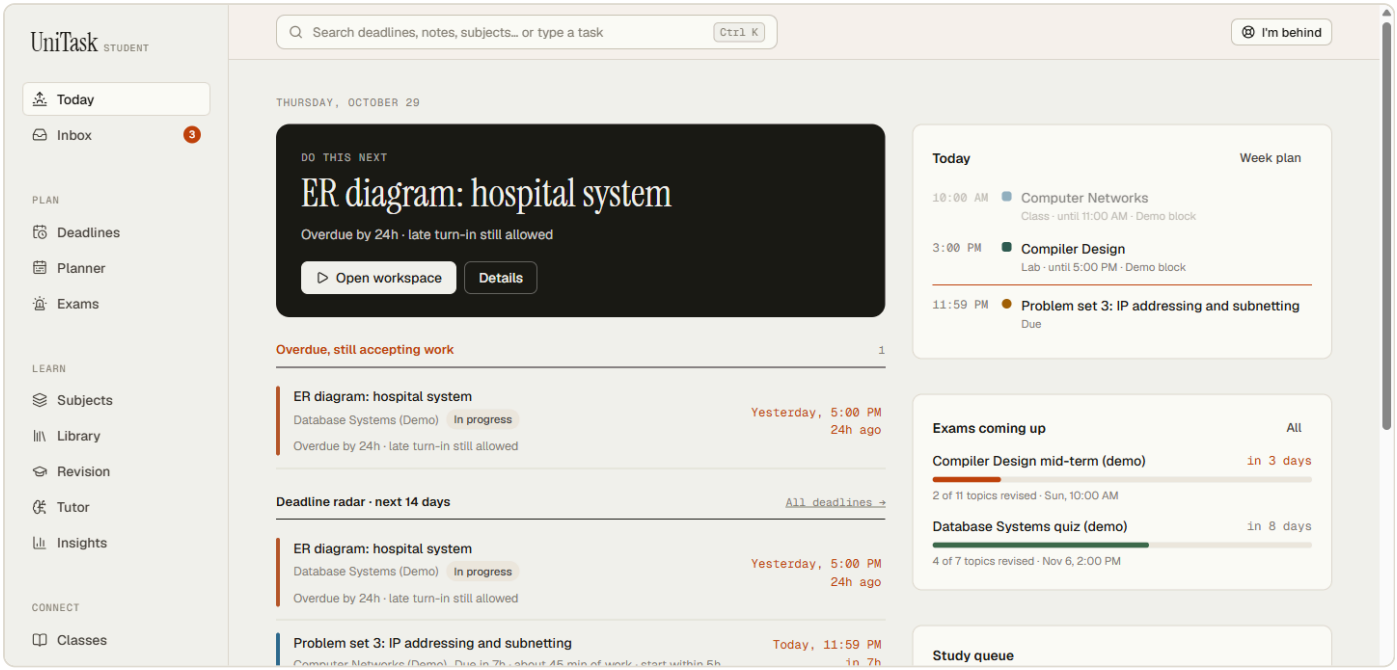
- No long questionnaire. A student reaches a useful screen in seconds.

SHOW IT IN THE DEMO

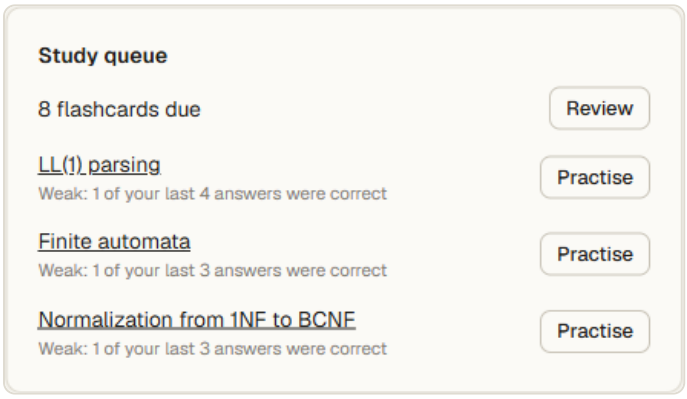
- Click **As a student** to open Riya's demo workspace.

04 Today: the academic command centre

The home screen answers one question: what should I do right now, and why?



Today on a wide screen



Study queue: flashcards due and weak topics to practise

WHAT IT SHOWS

- **Do this next**: one action with its reason, effort and time left, and a button to start.
- **Today**: classes, labs, exams and planned study in time order.
- **Exams coming up** with syllabus coverage, the **study queue**, and an honest weekly summary.
- **Deadline radar** for the next 14 days and a free-time planner for gaps between classes.

HOW IT HELPS THE STUDENT

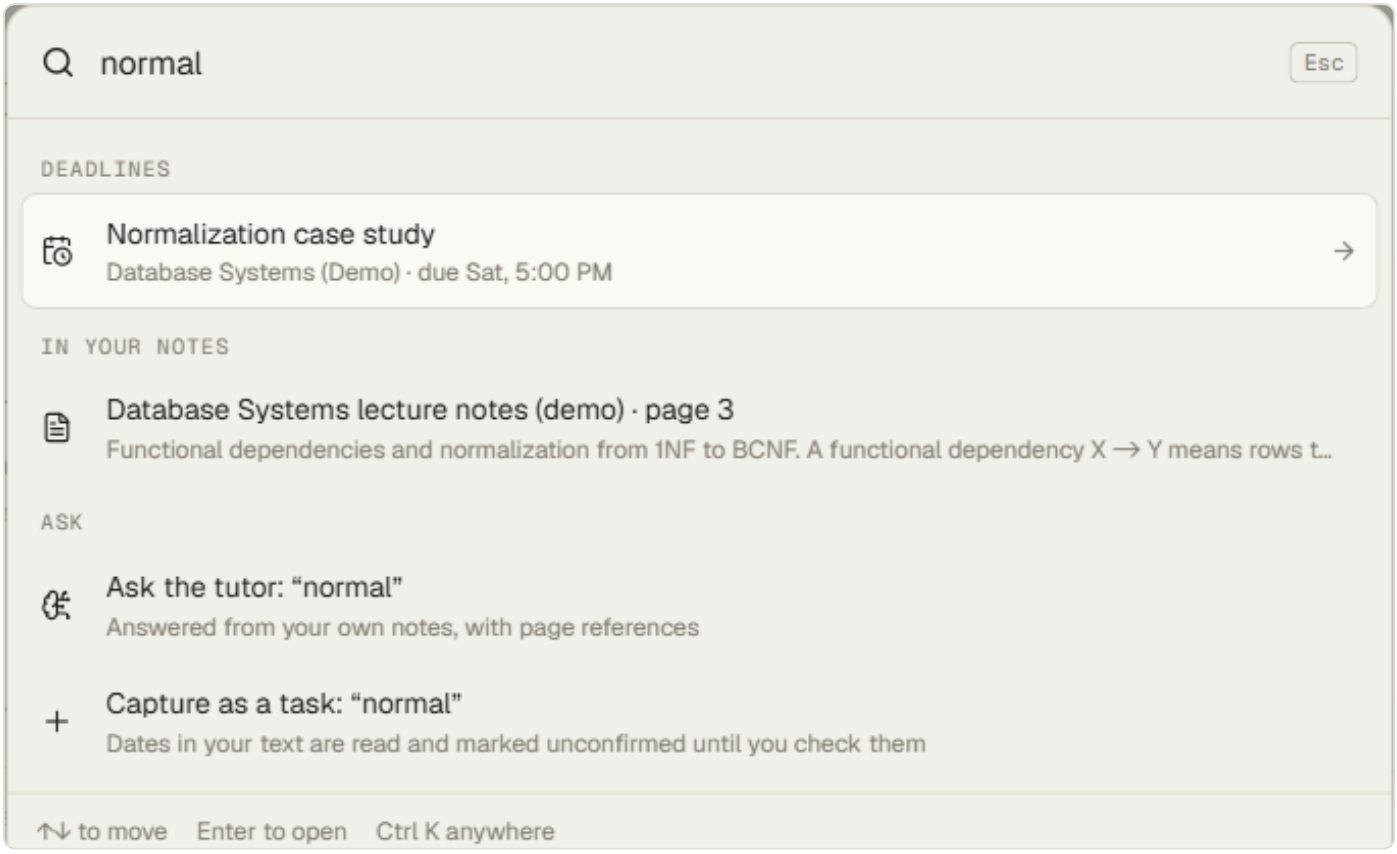
- Removes the “where do I start” paralysis.
- Work is ranked by work left against time left, so a 3-hour lab due tonight beats a 20-minute quiz due tomorrow.

SHOW IT IN THE DEMO

- Point at the dark card, read the reason, then click **Open workspace**.

05 Command bar (Ctrl K)

Search everything and capture anything from any screen.



Typing “normal” finds the deadline and the pages in the notes that discuss normalization

NEW IDEA Searches inside the text of the student's own notes and past papers, not just titles, and jumps to the exact page.

WHAT IT SHOWS

- Pages, deadlines, subjects and matching note pages as you type.
- **Ask the tutor** and **Capture as a task** for whatever you typed.

HOW IT HELPS THE STUDENT

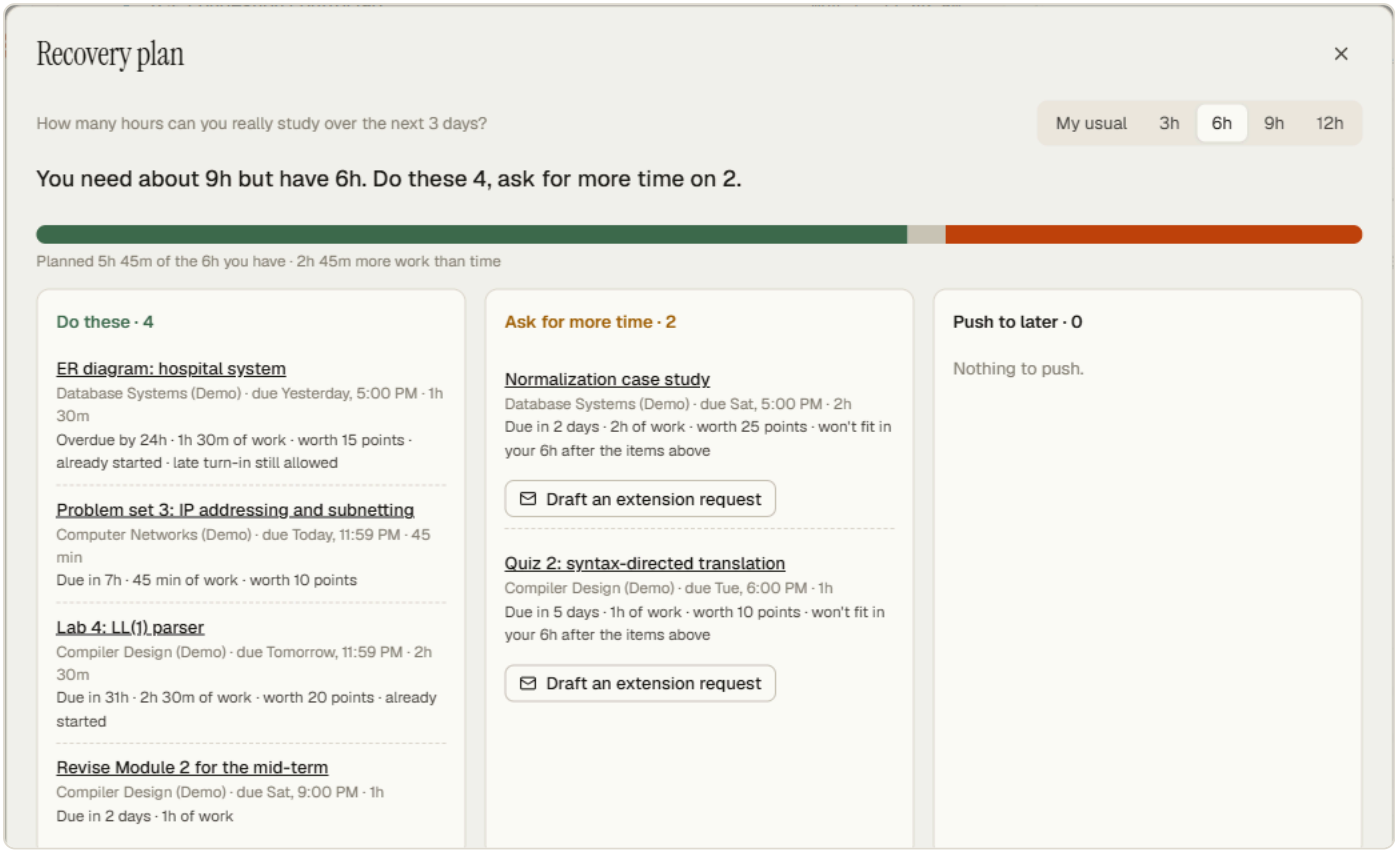
- Finding the one page that explained a concept takes seconds.
- Notices pasted from WhatsApp become tasks without filling in a form.

SHOW IT IN THE DEMO

- Press **Ctrl K**, type a topic, open the note page it finds.

06 “I'm behind”: recovery plan

When there is more work than time, UniTask makes a realistic triage for the next three days.



Do these, ask for more time, push to later, with a budget bar

> NEW IDEA Turns an overwhelming list into decisions, and drafts a polite extension request with a [reason] placeholder. It never sends anything.

WHAT IT SHOWS

- How many hours the work needs against the hours available.
- Three columns: **Do these**, **Ask for more time**, **Push to later**, each with a concrete reason.

HOW IT HELPS THE STUDENT

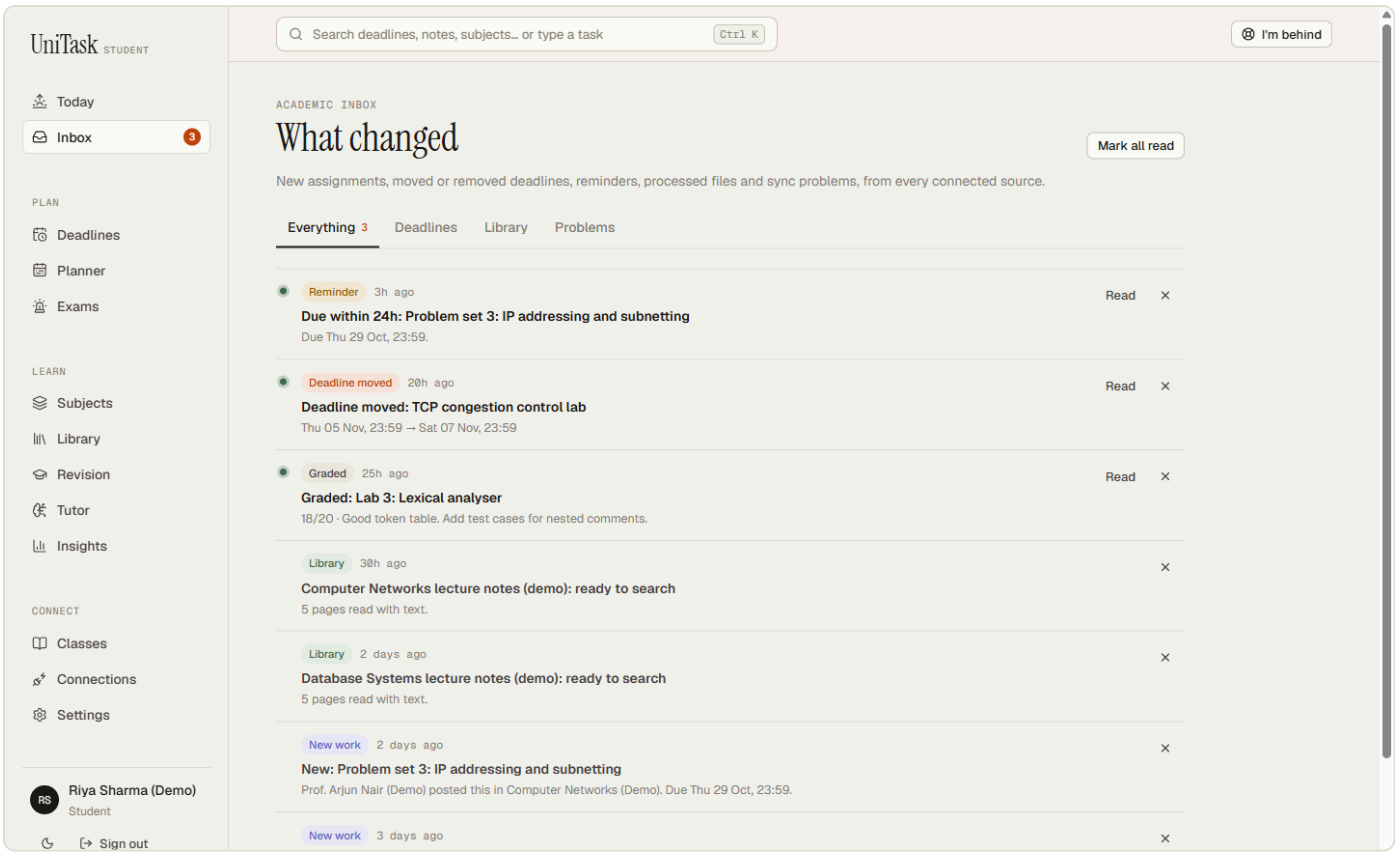
- Students stop trying to do everything and fail at all of it.
- Asking a teacher early, politely, is easier with a draft.

SHOW IT IN THE DEMO

- Click **I'm behind**, switch between 3h and 9h, and draft an extension email.

07 Academic inbox

Everything that changed, in one place: new work, moved deadlines, grades, ready materials and sync problems.



The academic inbox

WHAT IT SHOWS

- New assignments, deadline changes (old → new), returned grades, processed notes, reminders and integration errors.
- Filters, mark as read, dismiss, and a link to the source.

HOW IT HELPS THE STUDENT

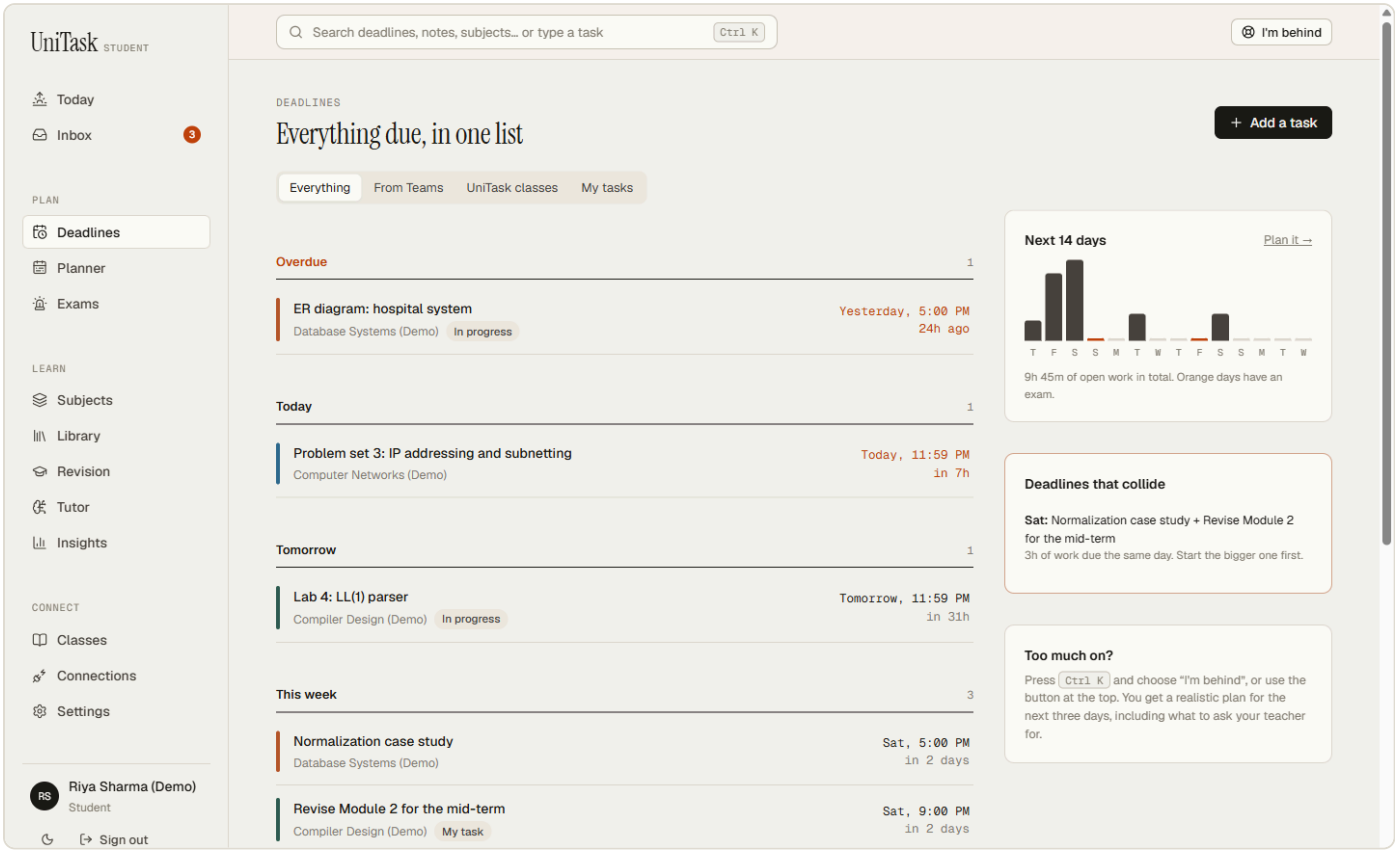
- A moved deadline is never missed. Each event appears exactly once, even if a sync runs twice.

SHOW IT IN THE DEMO

- Open the deadline-moved event and follow it to the assignment.

08 Deadlines

Every deadline from every source in one list, grouped by when it is due.



Deadlines with the 14-day workload strip and collision warnings

WHAT IT SHOWS

- Overdue (still accepting work), today, tomorrow, this week and later, with the source and status of each.
- A **14-day workload strip** (exam days in orange) and **deadlines that collide** on the same day.
- Dates read from pasted text are marked **unconfirmed** until the student checks them.

HOW IT HELPS THE STUDENT

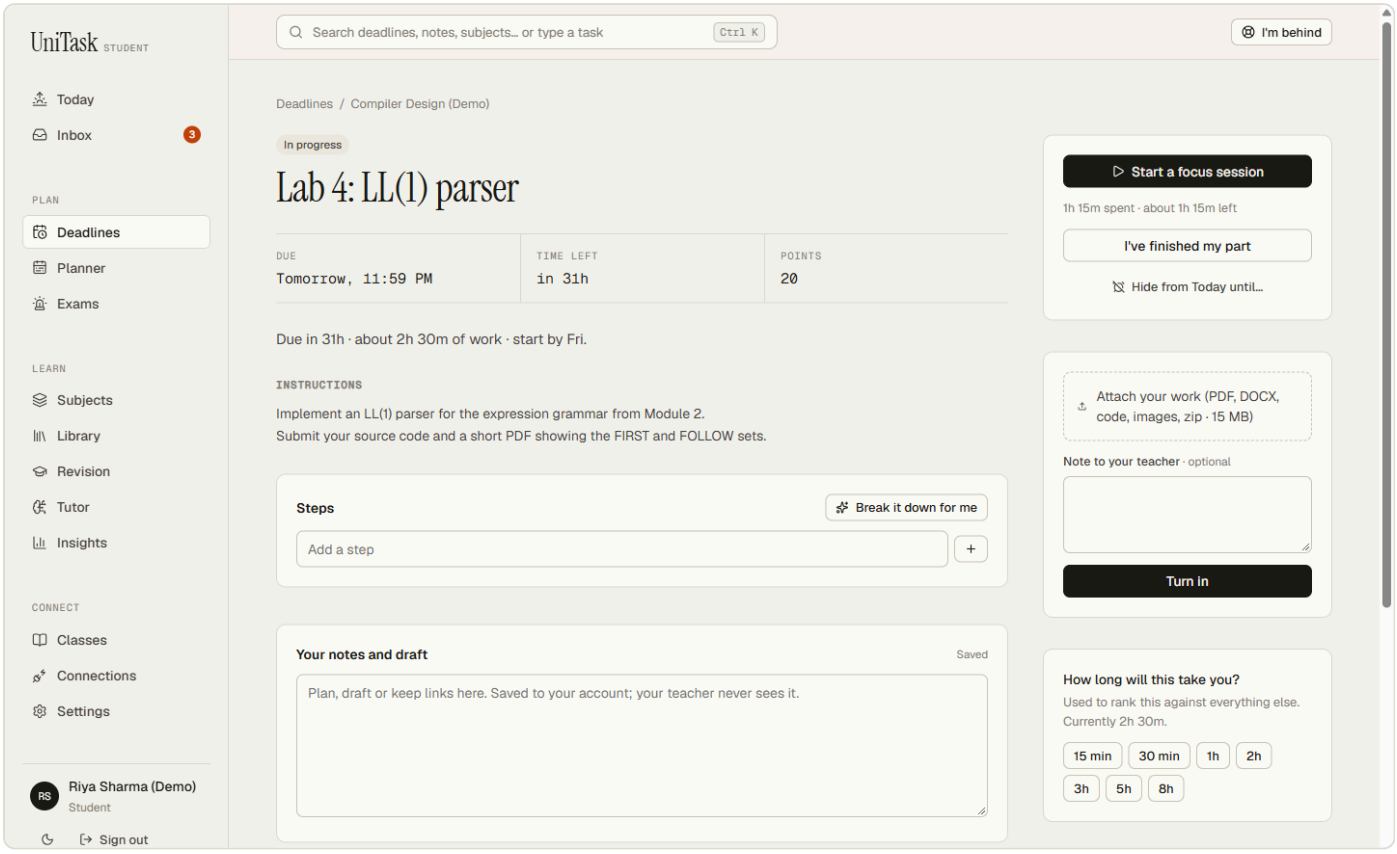
- Clashing deadlines are visible days in advance, while there is still time to start early.

SHOW IT IN THE DEMO

- Filter **From Teams** / **My tasks**, then point at the collision panel.

09 Assignment workspace

Each assignment opens as a workspace for actually doing it, not just its due date.



Assignment workspace: brief, steps, materials, notes and focus sessions

WHAT IT SHOWS

- The original brief, source link, due time and whether the date is verified.
- Steps (the AI can break the brief into subtasks), relevant pages from the student's notes, working notes and draft.
- Focus sessions, the submission state as the source reports it vs. the student's own **done**, and grades when returned.

HOW IT HELPS THE STUDENT

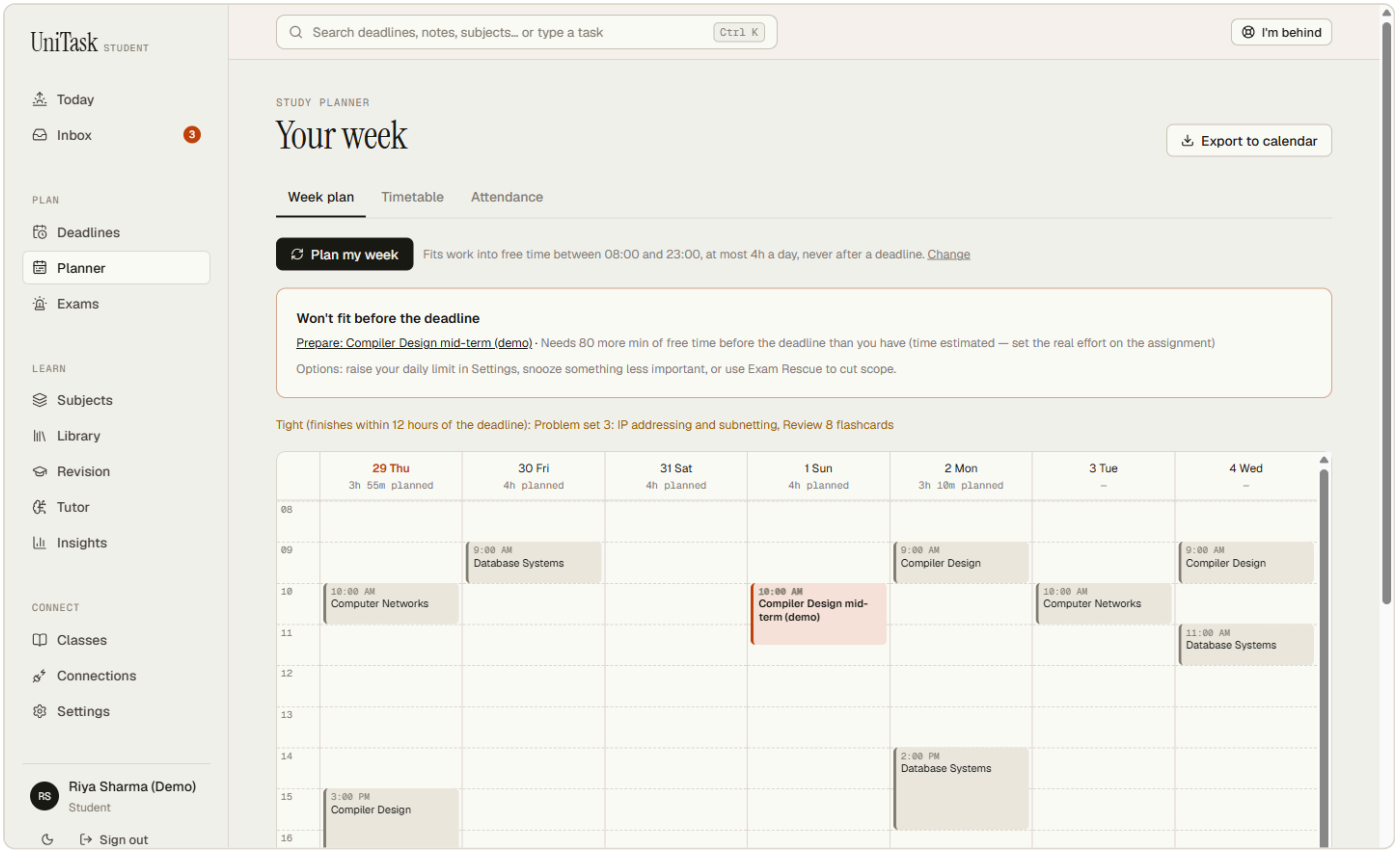
- The first step is obvious, so starting is easy.
- UniTask never pretends to submit to Teams: “submitted” only appears when Teams confirms it.

SHOW IT IN THE DEMO

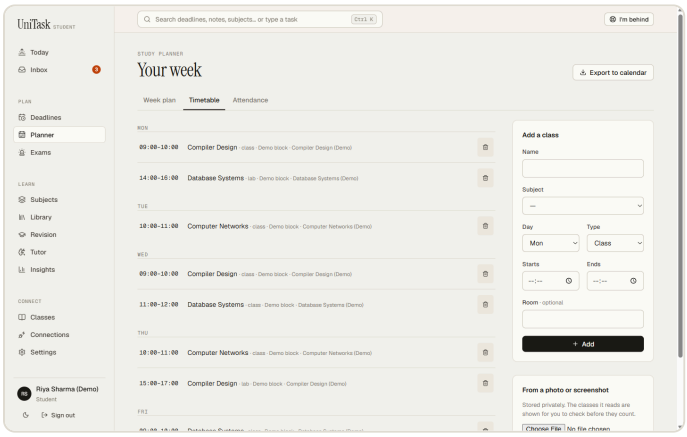
- Click **Break it down for me**, then **Start a focus session**.

10 Planner, timetable and attendance

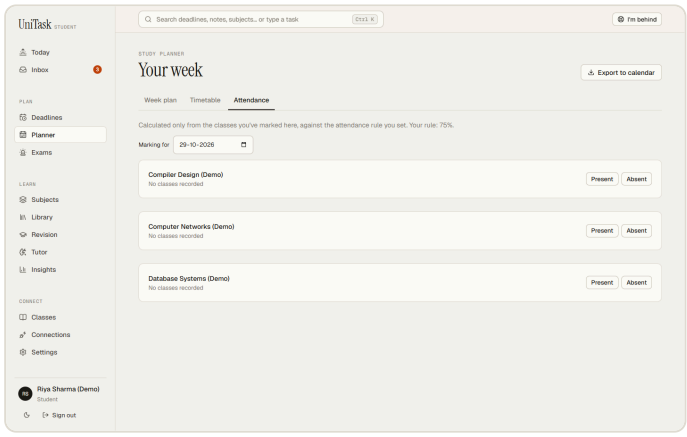
A week plan that fits work into real free time, around classes, labs and exams.



Week plan with classes, planned study and exams



Timetable: add classes or read them from a photo



Attendance against the student's own university rule

WHAT IT SHOWS

- **Plan my week**: study blocks between classes, capped per day, never after a deadline.
- **Won't fit before the deadline** and **tight** items listed honestly.
- Timetable from a photo (read, then confirmed), and attendance from real records.

HOW IT HELPS THE STUDENT

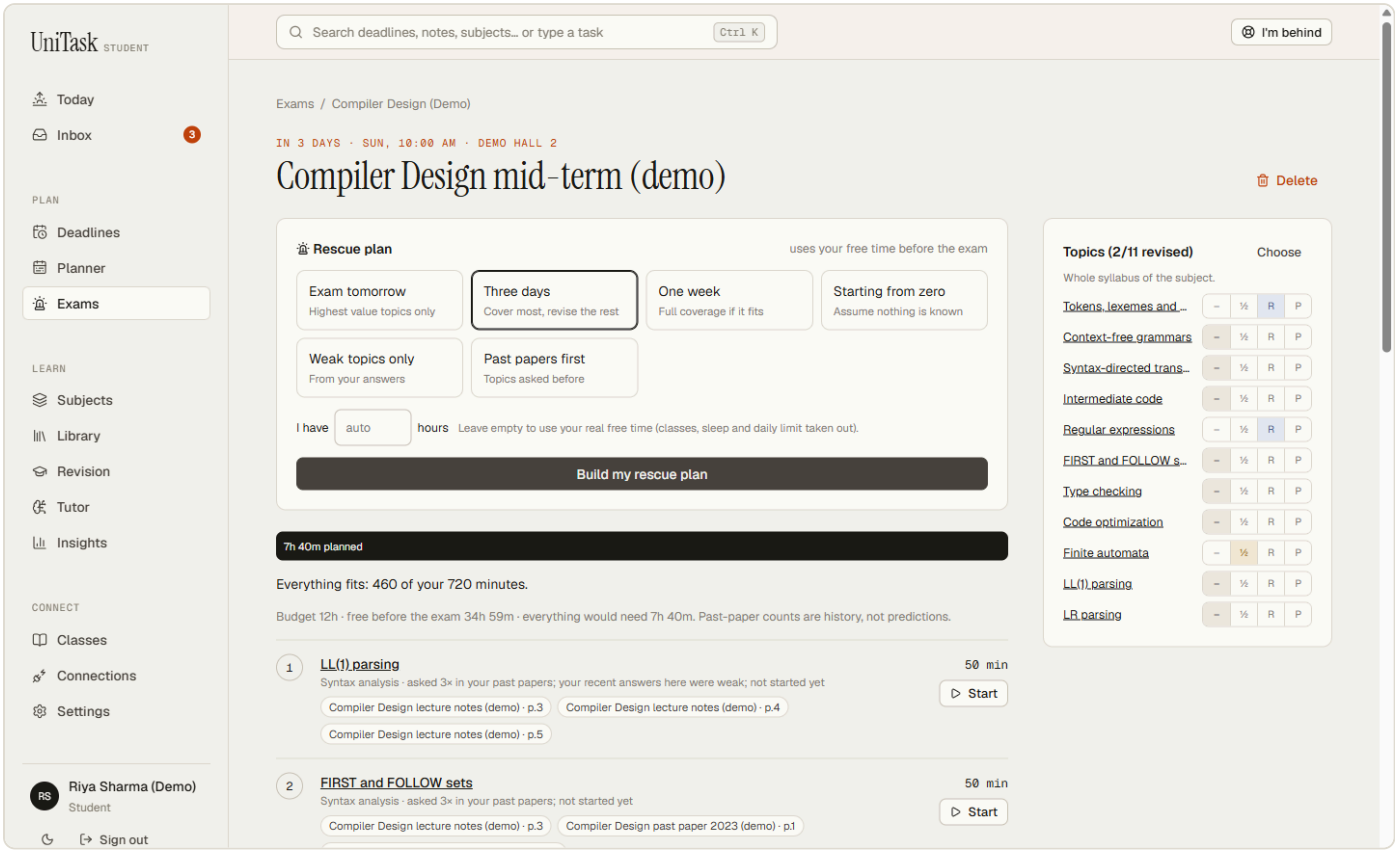
- The plan is feasible because it is calculated, not guessed. When a session is missed, **Replan from now** rebuilds the rest of the week.

SHOW IT IN THE DEMO

- Click **Plan my week** and point at the **Won't fit** box.

11 Exams and Exam Rescue

For each exam: the topics in scope, coverage so far, and a rescue plan that fits the time actually left.



Exam Rescue: modes, the time budget and the ordered plan with reasons

NEW IDEA Topic value = how often it appeared in the student's past papers + how many topics depend on it + what's not yet covered. Prerequisites come first.

WHAT IT SHOWS

- Modes: exam tomorrow, three days, one week, starting from zero, weak topics only, past papers first.
- A time-budget bar, the chosen topics in order with reasons, topics that don't fit and why, and the session schedule.

HOW IT HELPS THE STUDENT

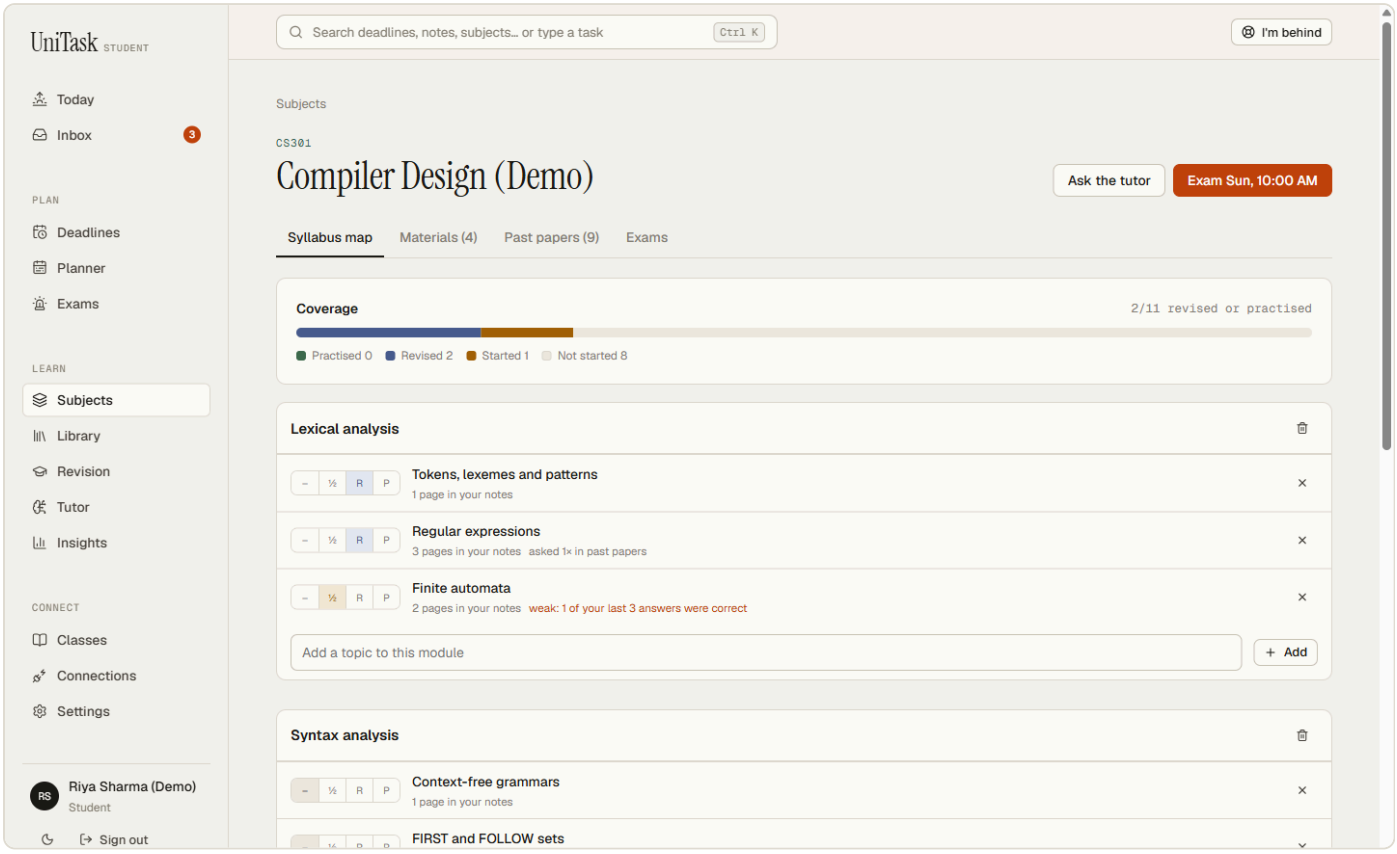
- The night before an exam goes to the topics that matter most for this student.
- No guarantees about marks; past frequency is shown as history.

SHOW IT IN THE DEMO

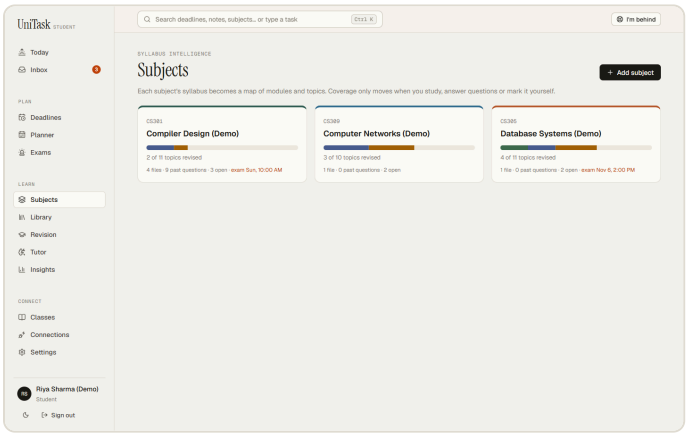
- Switch from **One week** to **Exam tomorrow** and watch the plan shrink, with reasons.

12 Subjects and the syllabus map

Each subject's syllabus becomes a map of modules and topics with real coverage.



Syllabus map: modules, topics, coverage and linked pages



All subjects with coverage and upcoming exams

WHAT IT SHOWS

- Syllabus read from an uploaded PDF or photo, then confirmed by the student.
- Coverage per topic: not started, in progress, revised, practised.
- Prerequisites (AI suggestions stay marked until accepted) and where the notes cover each topic.

HOW IT HELPS THE STUDENT

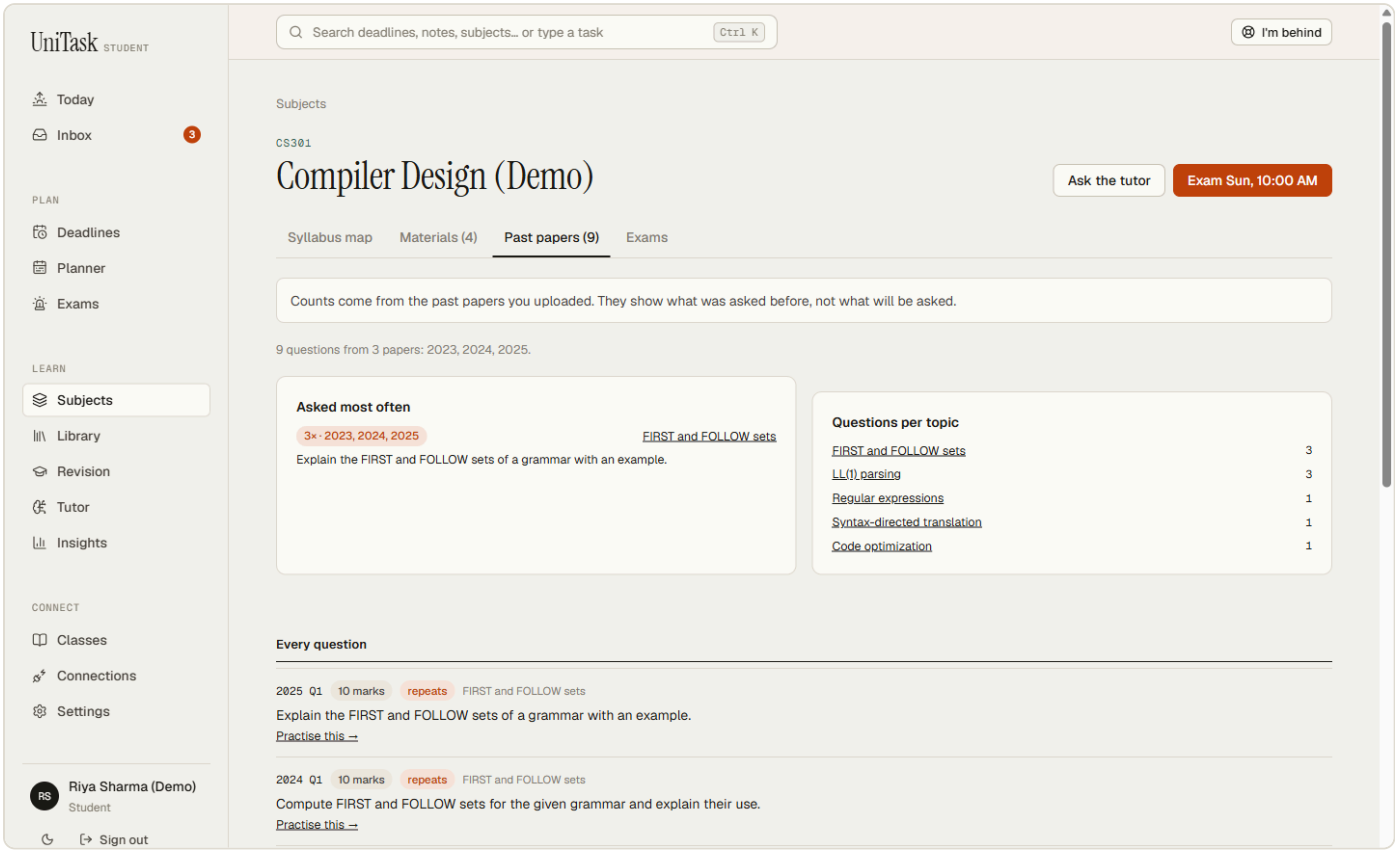
- Coverage only moves on evidence: study sessions, correct answers, or the student's own mark. No fake mastery percentages.

SHOW IT IN THE DEMO

- Open Compiler Design and point at a topic with linked note pages.

13 Past-paper intelligence

Upload previous-year papers. UniTask extracts the questions, maps them to topics and shows what repeats.



Questions per topic, most-asked patterns and every question with its year

WHAT IT SHOWS

- Questions kept word for word with year, marks and page.
- Counts per topic and repeated question patterns across years.

HOW IT HELPS THE STUDENT

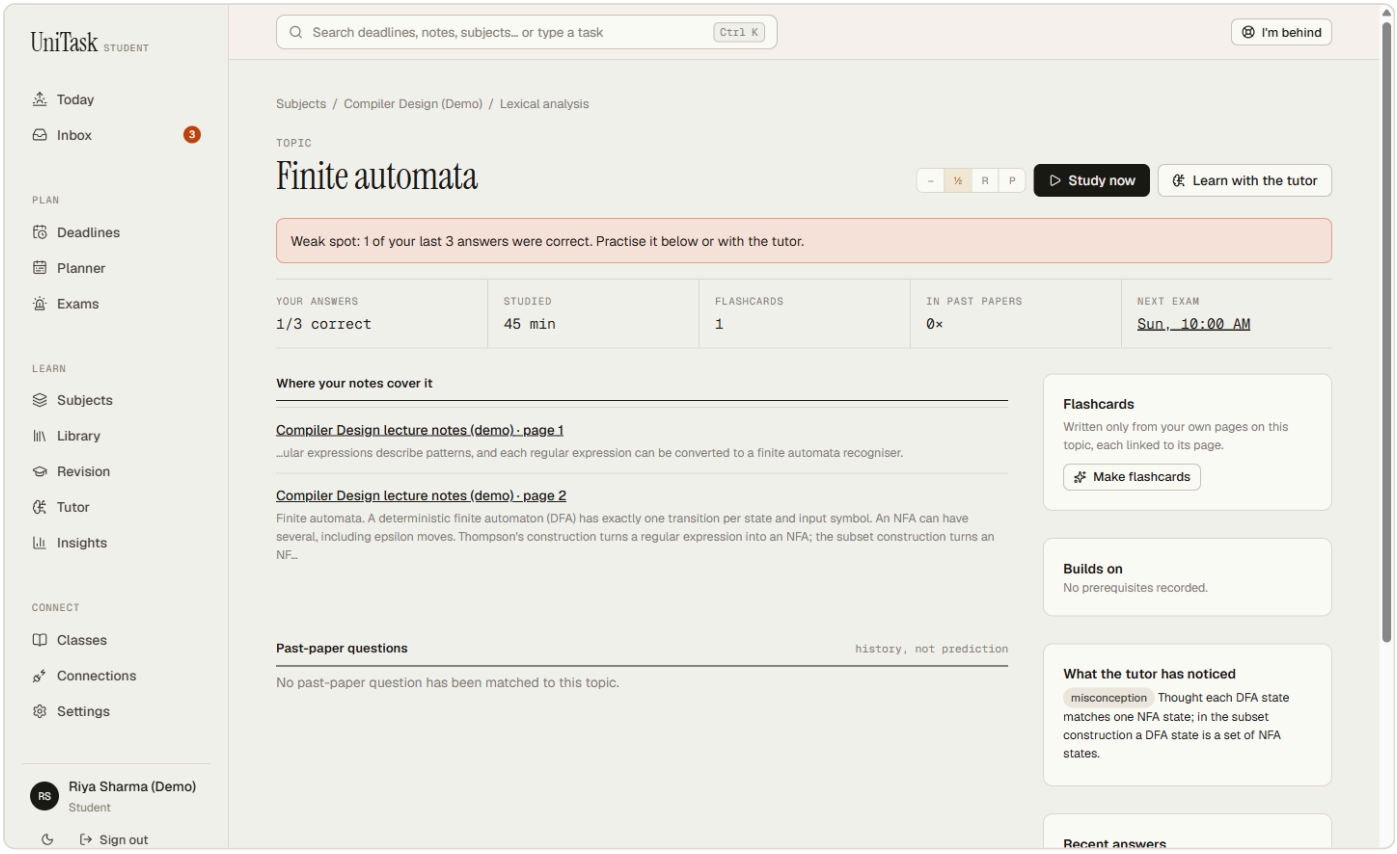
- Students see which topics their university actually asks. This feeds Exam Rescue.

SHOW IT IN THE DEMO

- Show **Asked most often**, then the same question in two different years.

14 Topic page

Everything about one topic in one place.



A topic: notes pages, past questions, flashcards, recent answers and what the tutor noticed

WHAT IT SHOWS

- Where the notes cover it, past-paper questions, flashcards, recent answers, prerequisites, and what the tutor has noticed.

HOW IT HELPS THE STUDENT

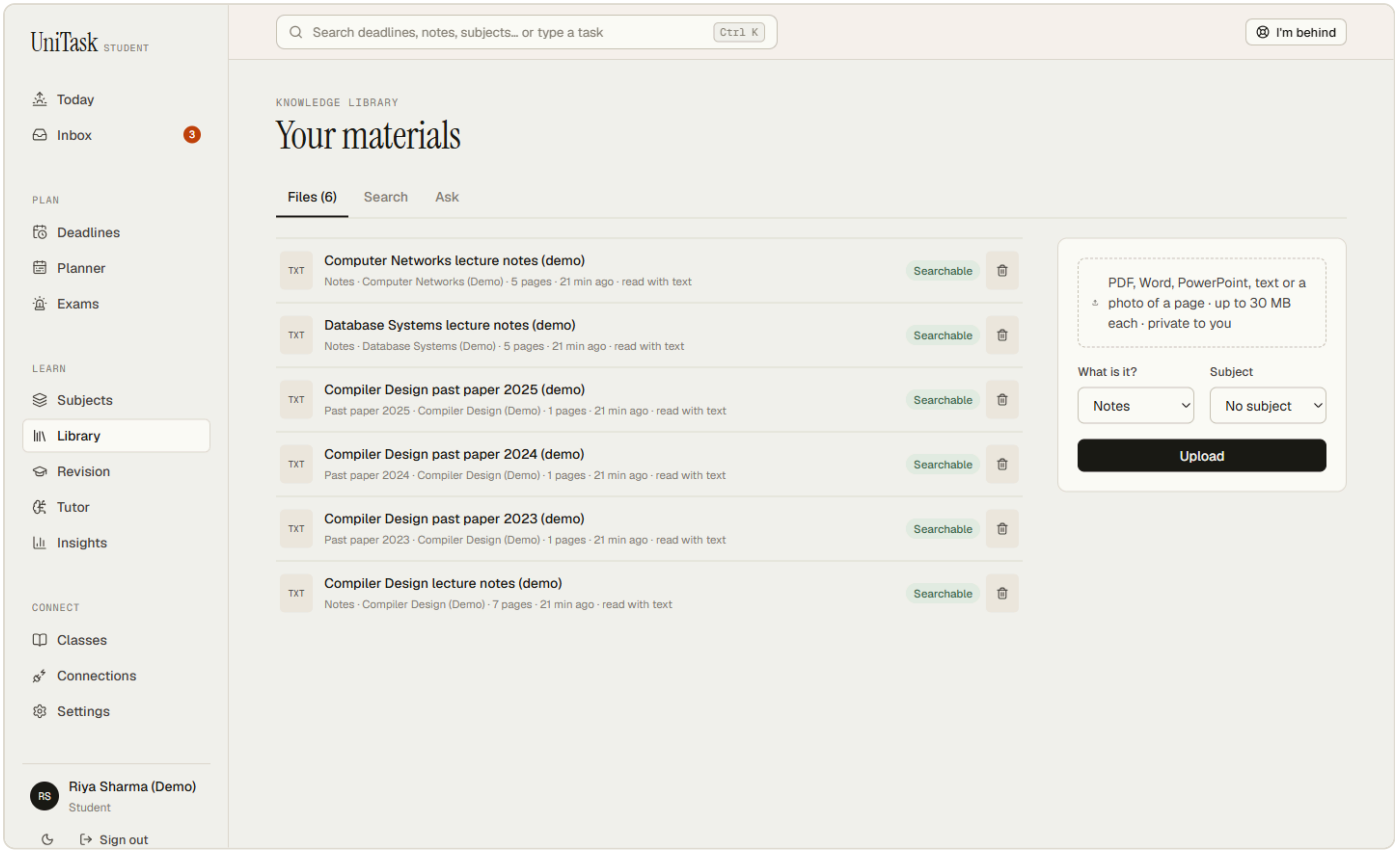
- One click from "I'm weak at this" to studying it with the right material.

SHOW IT IN THE DEMO

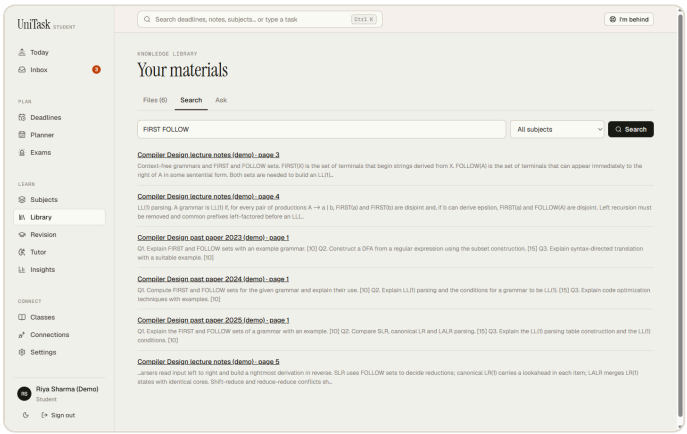
- Click **Learn with the tutor**.

15 Library: notes, slides and papers

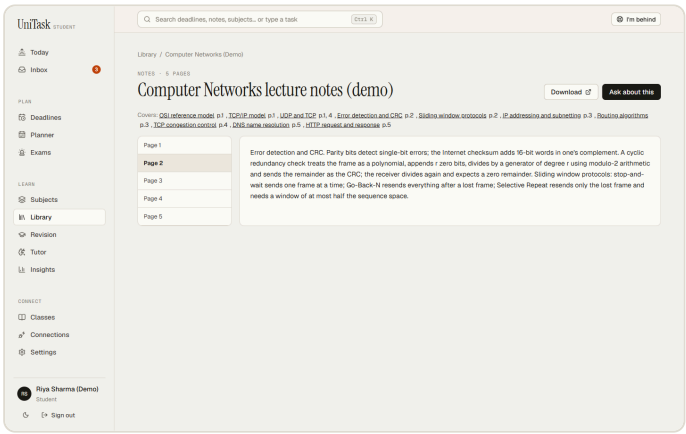
Private storage for academic material, read page by page so it can be searched and cited.



The library, with processing status per file



Search results with excerpts and page numbers



Reading a material page by page

WHAT IT SHOWS

- Uploads of PDF, Word, PowerPoint, images and text, stored privately on Amazon S3.
- Text extraction (scans via OCR), duplicate detection and processing status.
- Search with excerpts and page references.

HOW IT HELPS THE STUDENT

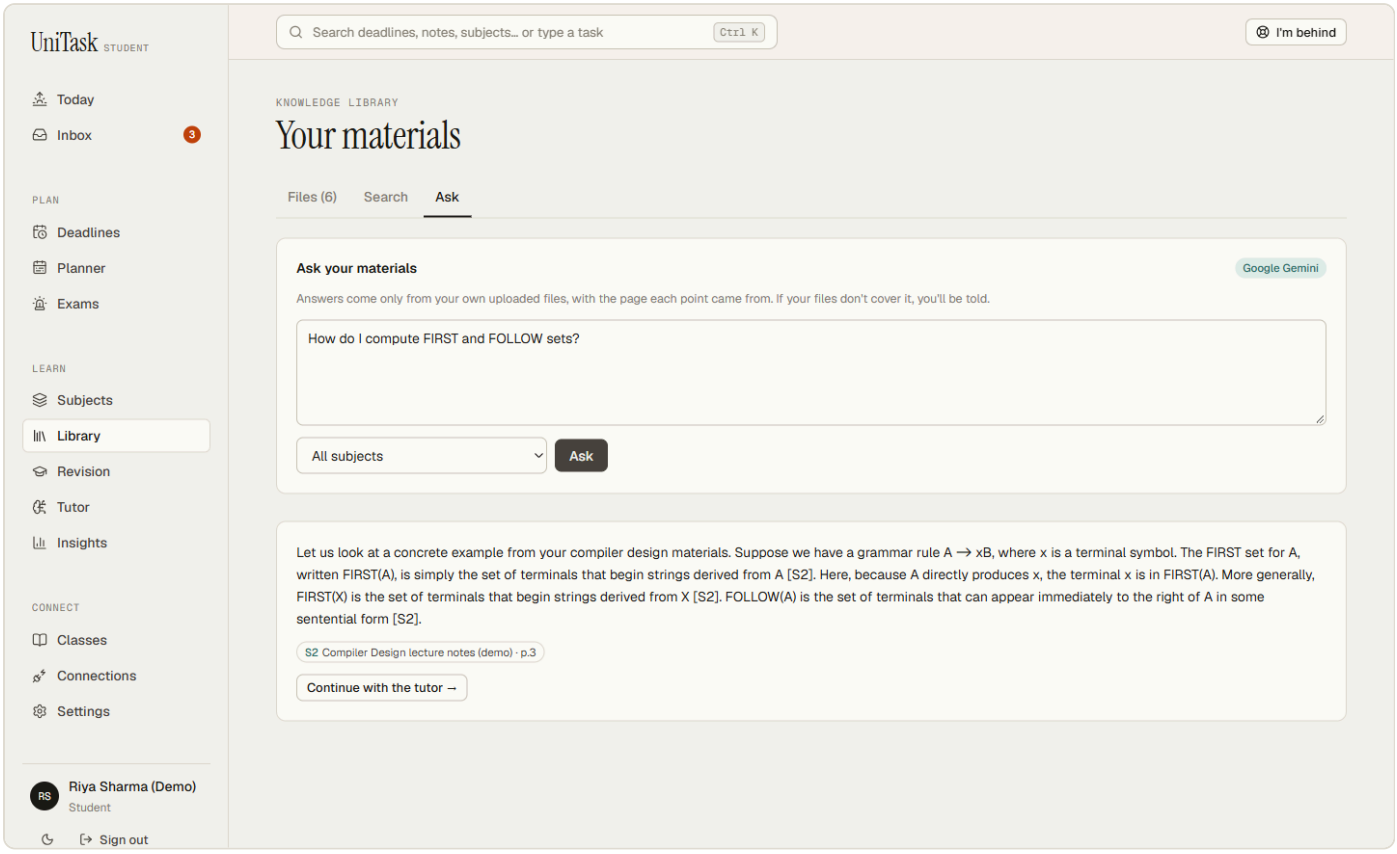
- "Where did the lecturer explain LR parsing?" has an answer with a page number.

SHOW IT IN THE DEMO

- Search **FIRST FOLLOW** and open the result on its page.

16 Ask your notes, with page citations

Questions are answered from the student's own material, citing file and page.



A grounded answer: every claim links to the page it came from

> NEW IDEA Citations are validated against the pages the AI was actually given. Anything else is removed. If the notes don't cover it, the answer says so.

WHAT IT SHOWS

- The answer, numbered citations [S1], [S2] and the excerpt behind each.

HOW IT HELPS THE STUDENT

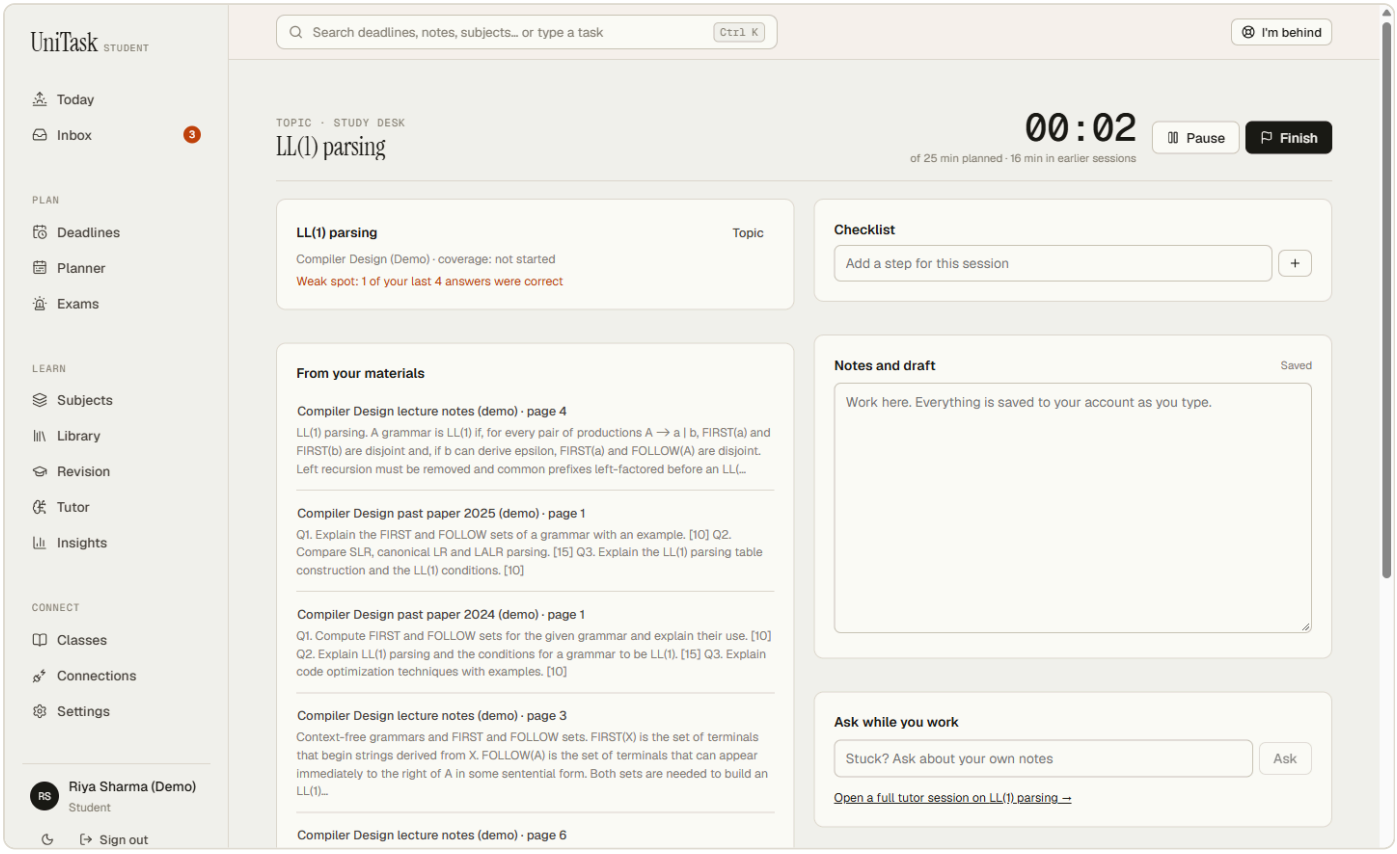
- Answers match the syllabus and the lecturer's notation, not a random website.

SHOW IT IN THE DEMO

- Ask **How do I compute FIRST and FOLLOW sets?** and click a citation.

17 Study desk

A distraction-free workspace for one focused session.



Study desk: timer, checklist, material panel, tutor and notes

WHAT IT SHOWS

- Session timer with pause and resume, a step checklist, the relevant pages, due flashcards and an **ask while you work** tutor.
- Scratch notes and an exact resume point.

HOW IT HELPS THE STUDENT

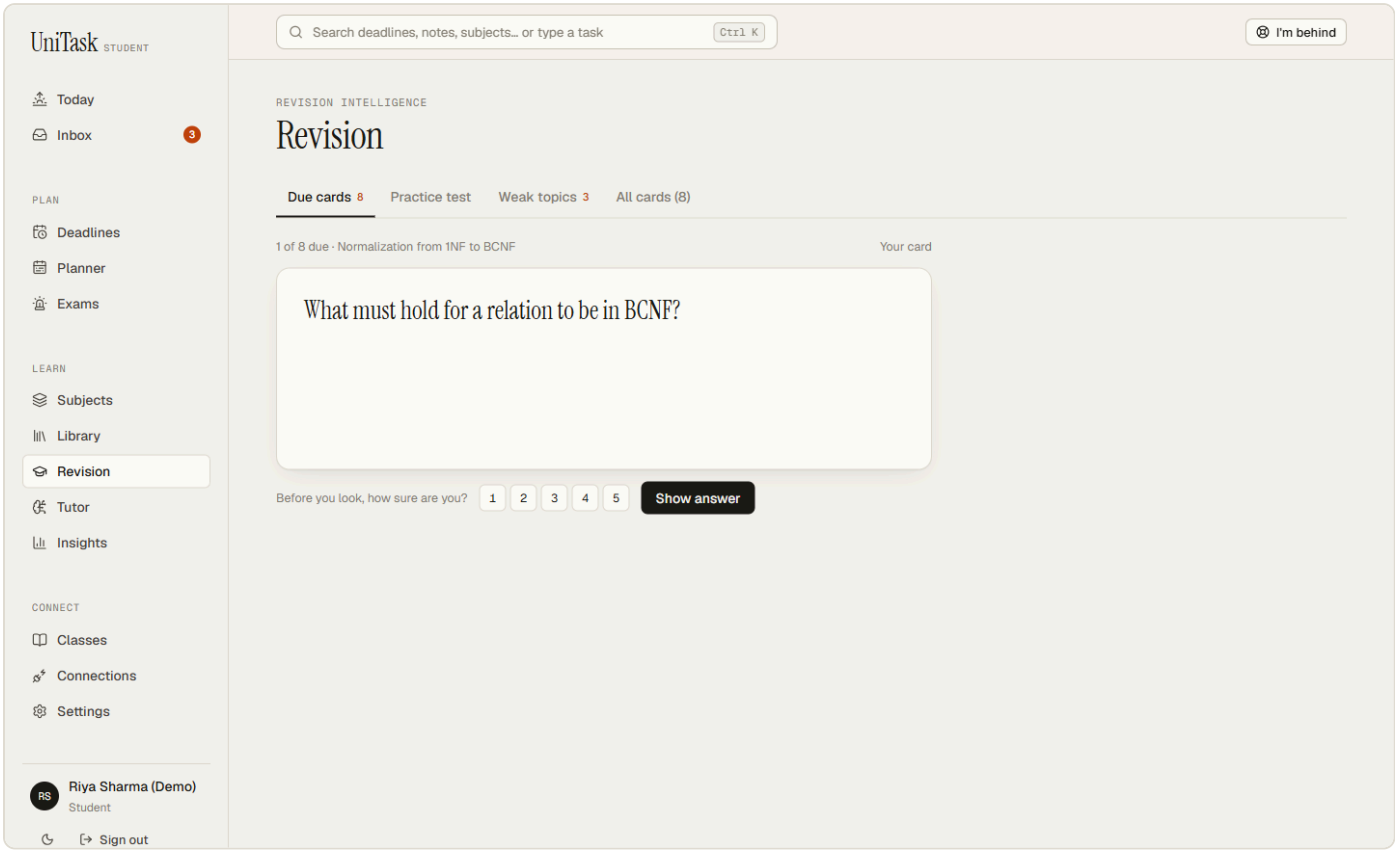
- Sessions are recorded as real evidence, never just for opening a page.
- Stopping and continuing later picks up exactly where the student left off.

SHOW IT IN THE DEMO

- Start a session from Today, tick a step, pause and resume.

18 Revision and weak topics

Spaced repetition and weak-topic detection from the student's real answers.



Due cards with again / hard / good / easy, weak topics and practice tests

WHAT IT SHOWS

- Due flashcards (from notes, by hand, or from mistakes the tutor caught).
- Weak topics with the evidence, e.g. “1 of your last 3 answers were correct”.
- Practice tests from past papers, graded with feedback.

HOW IT HELPS THE STUDENT

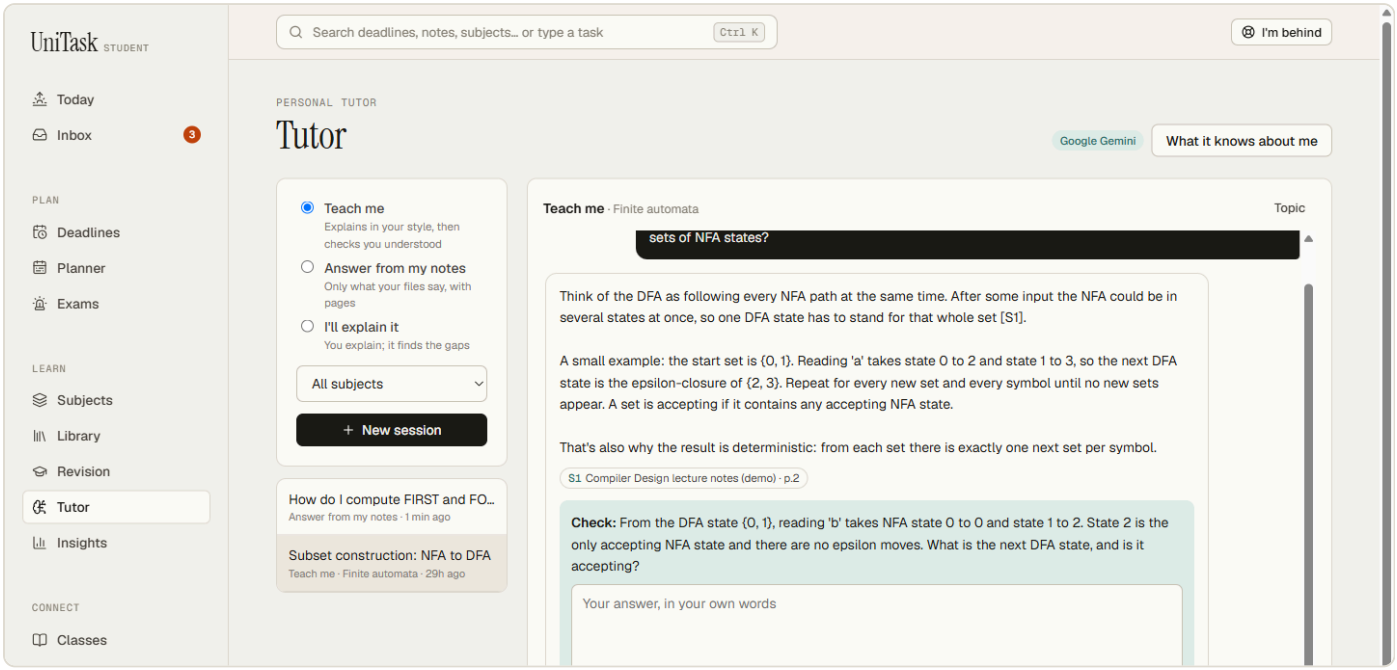
- Revision goes where it is needed, at the right time, with a reason the student can check.

SHOW IT IN THE DEMO

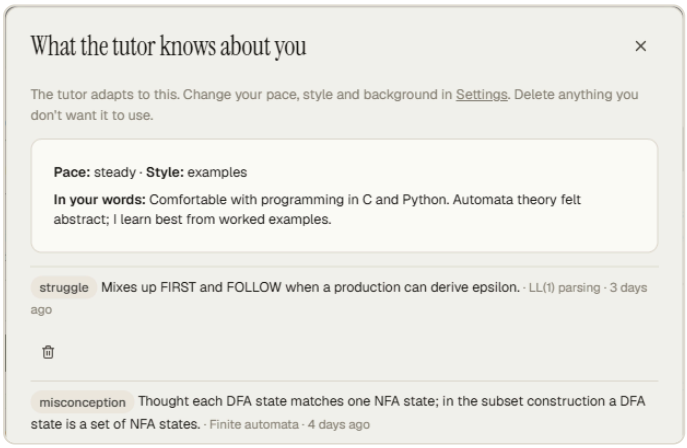
- Review a card and grade it **Good**; show its next date.

19 Personal tutor

A tutor that teaches this particular student at their pace and checks they understood.



A tutoring session: grounded explanation with citations, then a check question



What the tutor knows about the student, which they can delete

NEW IDEA Personalised by evidence: pace, style, background, past answers and remembered misconceptions shape every reply. Three modes: **Teach me**, **Answer from my notes**, **I'll explain it** (the student explains; the tutor finds the gaps).

WHAT IT SHOWS

- Explanations grounded in the student's pages, with citations.
- One check question per concept; answers are graded and recorded.
- A transparent memory the student controls.

HOW IT HELPS THE STUDENT

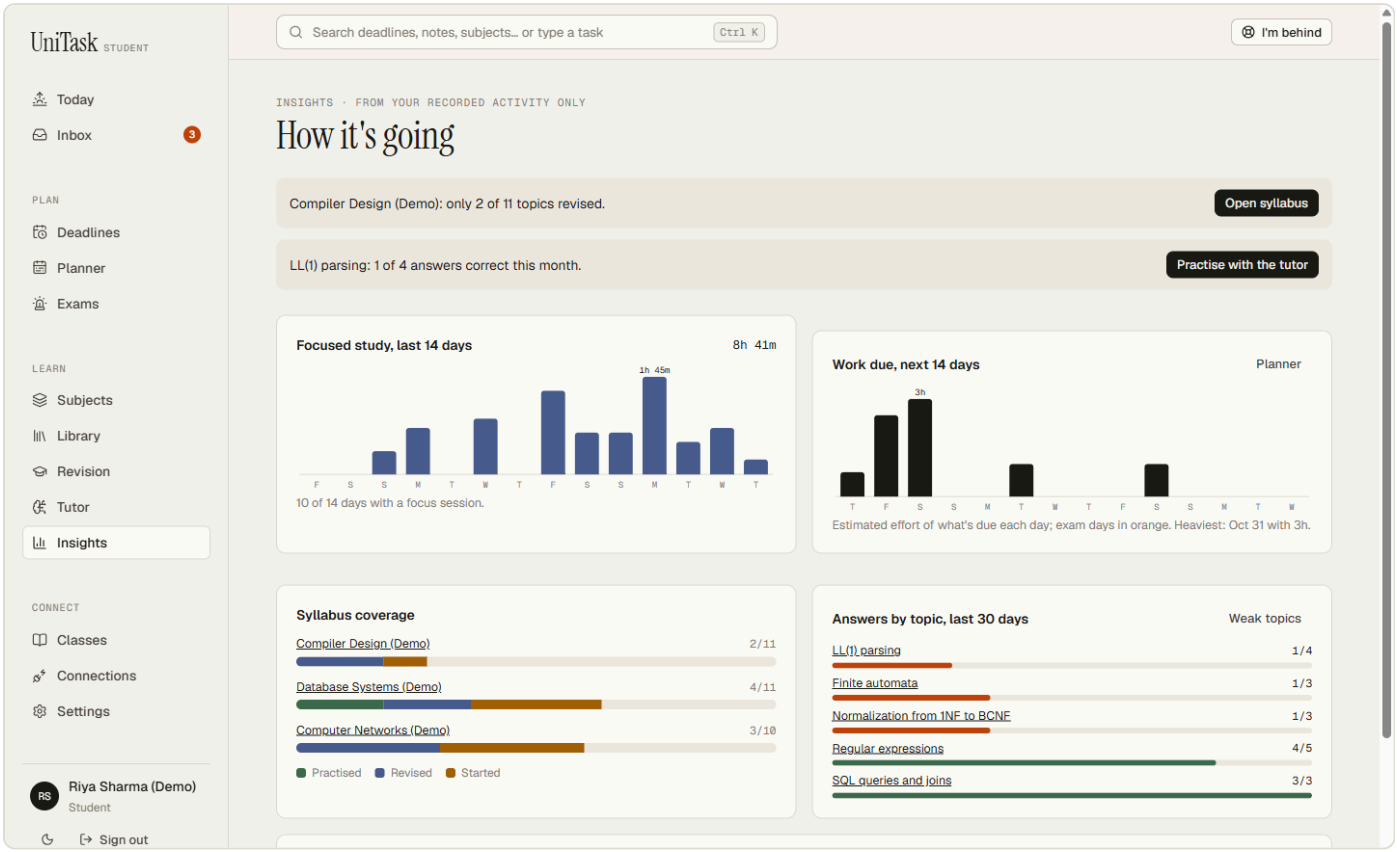
- A weak student gets foundations and smaller steps; a fast one skips what they already showed they know.
- Mistakes become flashcards automatically, so they get fixed.

SHOW IT IN THE DEMO

- Open the Finite automata session, answer the check question, then open **What it knows about me**.

20 Insights

Progress from recorded activity only: no invented productivity scores.



Focus time, upcoming workload, coverage, answers by topic and on-time rate

WHAT IT SHOWS

- Focused study over 14 days, work due over the next 14 days (exam days highlighted), syllabus coverage, accuracy by topic, finished-on-time rate.
- Suggested actions linked to the evidence behind them.

HOW IT HELPS THE STUDENT

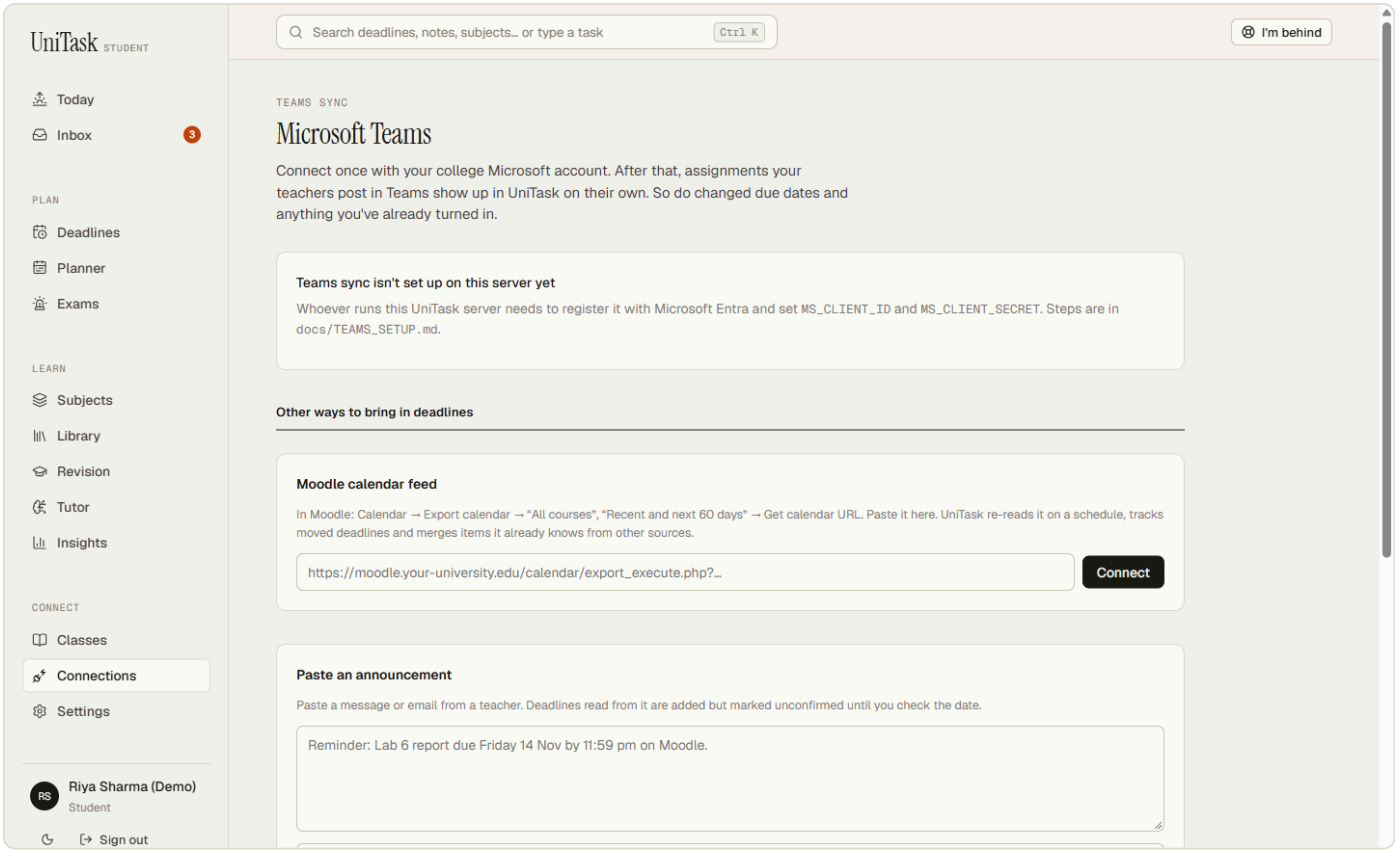
- The student sees where effort actually went and what to fix next.

SHOW IT IN THE DEMO

- Point at the heaviest day ahead and click **Replan the week**.

21 Connections: deadlines arrive by themselves

UniTask brings work in from where it already lives, so students don't retype it.



Teams, Moodle calendar feed, VTOP extension, paste a notice, file import

WHAT IT SHOWS

- **Microsoft Teams** via Microsoft Graph: read-only, the student signs in on Microsoft's page, sync every 30 minutes.
- **Moodle calendar feed**, **CSV/JSON/ICS import**, **paste an announcement**, and a **VTOP browser extension**.
- Cross-platform de-duplication, change history and clear error states. Nothing is faked when a connection is missing.

HOW IT HELPS THE STUDENT

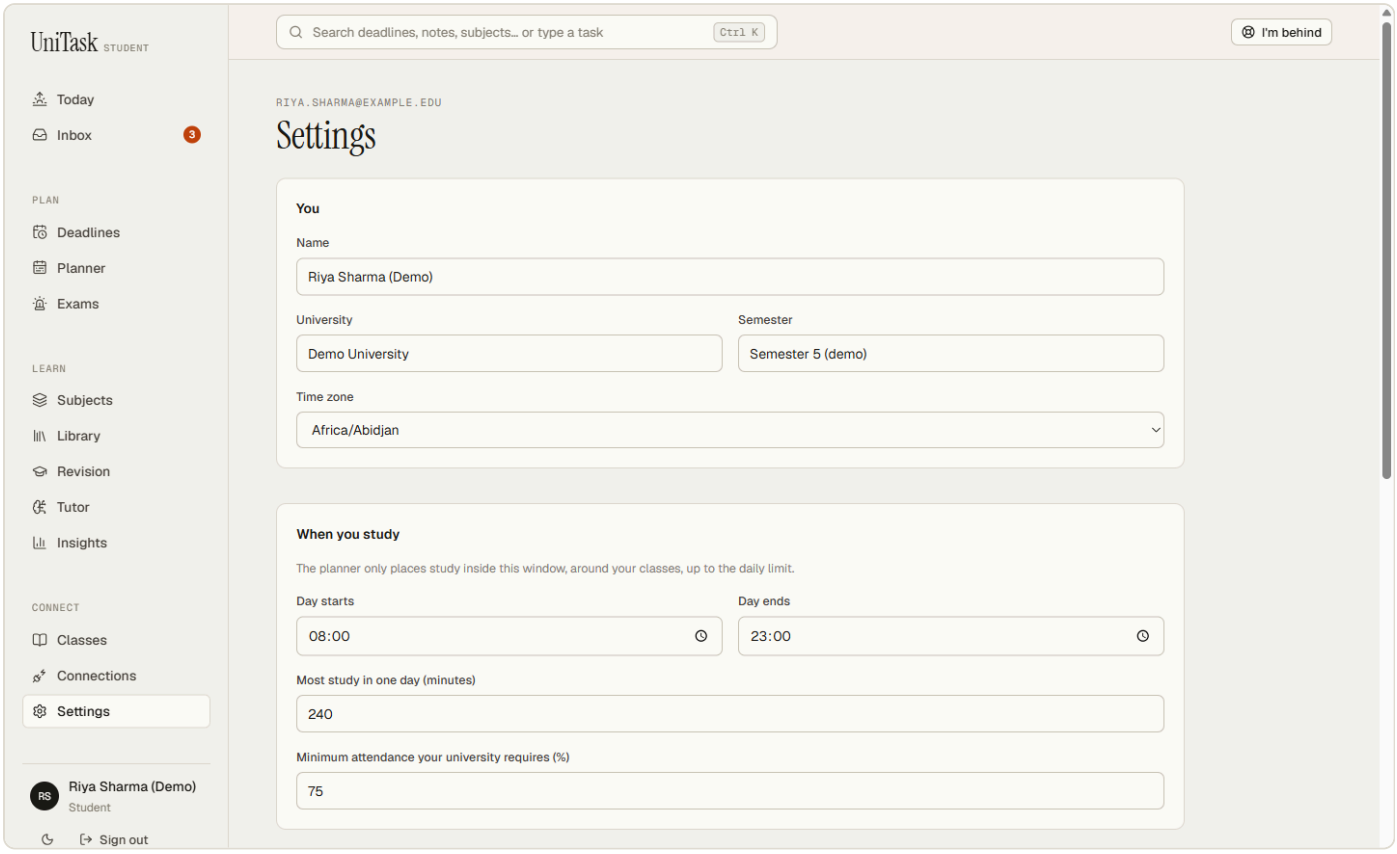
- No more copying deadlines by hand. Moved deadlines are caught automatically.

SHOW IT IN THE DEMO

- Show the Teams card and the paste-a-notice box.

22 Settings: how you learn

A short profile that the planner and the tutor actually use.



Study hours, pace, explanation style and background, data controls

WHAT IT SHOWS

- Study window and daily limit (planner).
- Pace, explanation style and background in the student's words (tutor).
- Attendance rule, AI and service status, data import and deletion.

HOW IT HELPS THE STUDENT

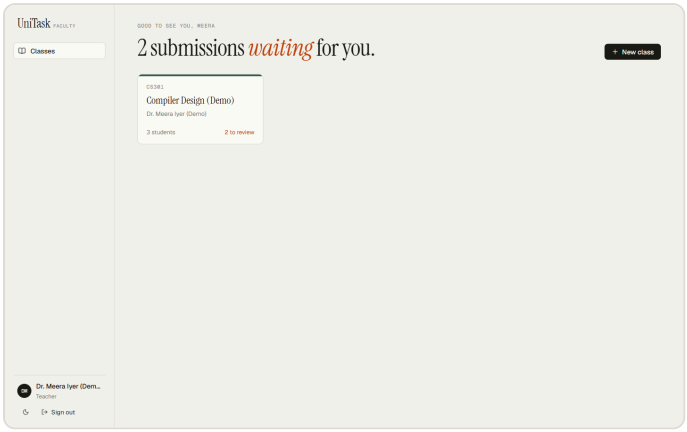
- Two minutes of setup makes every plan and every explanation fit the student.

SHOW IT IN THE DEMO

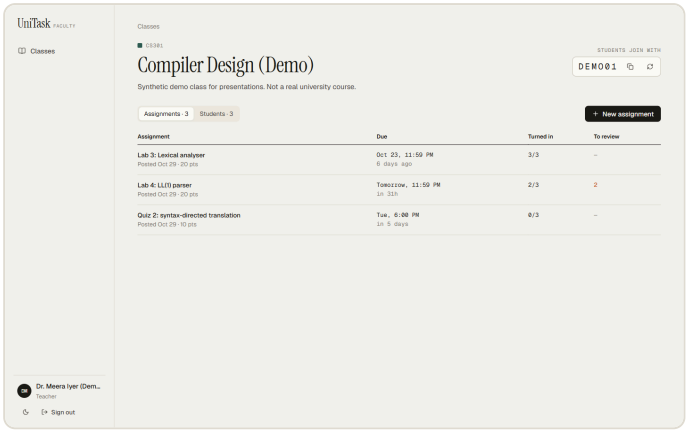
- Change the explanation style to **Step by step** and ask the tutor again.

23 Classes and the teacher side

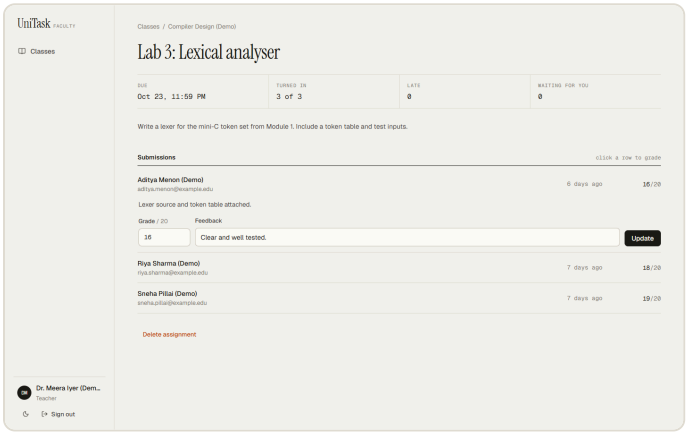
Teachers run classes like Teams assignments; students see the work inside their own planning.



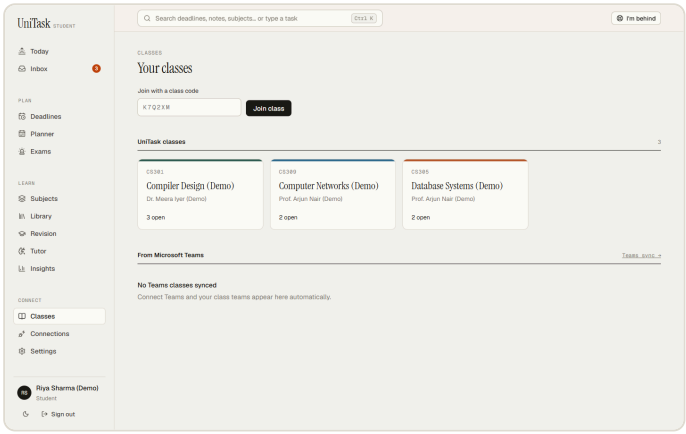
Teacher: classes and submissions waiting



Teacher: a class, its join code and assignments



Teacher: grading a submission



Student: joined classes and Teams classes

WHAT IT SHOWS

- Create a class and share a 6-letter join code. Post assignments with handouts and points.
- See who turned in (and who was late), download submissions, grade, give feedback and return.

HOW IT HELPS THE STUDENT

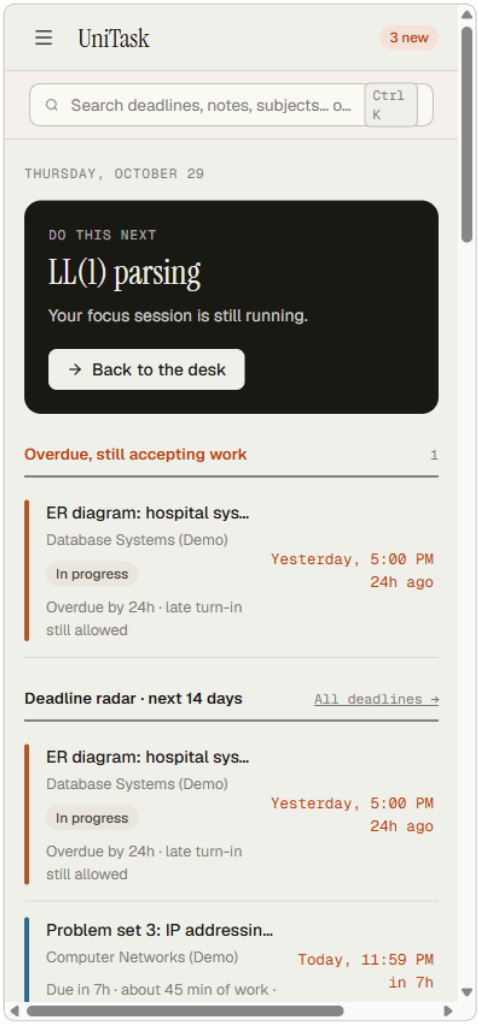
- Teacher-posted work drops straight into each student's Today, planner and recovery plan.

SHOW IT IN THE DEMO

- As the teacher, return a graded submission. As the student, the grade appears in the inbox.

24 On a phone

The whole app works on a phone, which is where students actually check deadlines.



Today on a phone

WHAT IT SHOWS

- Responsive layout, a slide-out menu and an unread badge.

HOW IT HELPS THE STUDENT

- The next action is one tap away between classes.

SHOW IT IN THE DEMO

- Resize the browser or open the deployed URL on a phone.

25 How it is built (AWS)

Part	What runs it
Web app and API	One container on **Amazon ECS** (Fargate, HTTPS load balancer) serving the React site and the FastAPI API.
Database	**Amazon RDS for PostgreSQL**, with versioned Alembic migrations. SQLite for local development.
Files	**Amazon S3**: private bucket, encrypted, with 5-minute presigned download links.
AI	**Amazon Bedrock** when enabled, with **Google Gemini** as fallback. Used to explain, structure and tutor; never to decide deadlines or feasibility.
OCR	**Amazon Textract** for scanned pages and photos (or the vision model).
Reminders and email	**Amazon EventBridge** schedules → reminder job → **Amazon SES** email digest.
Accounts	Local accounts or **Amazon Cognito** user pool.
Monitoring	**Amazon CloudWatch** logs from the container. Budget alerts at $5 / $15 / $30.
Integrations	**Microsoft Graph** (Teams Education, read-only, delegated), Moodle iCalendar feeds, VTOP extension.

Five-minute demo script

1. **Problem (30 s).** Deadlines are scattered and students study the wrong thing at the wrong time. A dashboard that counts deadlines doesn't help.
2. **Today (1 min).** Do this next, the reason and the day's timeline. Open the workspace and break the assignment into steps.
3. **Ctrl K (30 s).** Type a topic and jump to the exact page in the notes. Type a notice to capture a task.
4. **I'm behind (45 s).** The triage, and the drafted extension email.
5. **Exam Rescue (1 min).** Switch to “Exam tomorrow”: past-paper frequency and prerequisites decide the order; topics that don't fit are listed with reasons.
6. **Ask your notes (45 s).** A cited answer; click a citation to open the page.
7. **Tutor (1 min).** A check question and what the tutor remembers about the student.
8. **Teacher (30 s).** Grade and return. The student sees it in the inbox.
9. **Close (15 s).** Runs on AWS: ECS, RDS, S3, Bedrock/Textract, EventBridge, SES, CloudWatch.

26 Honest answers to hard questions

Isn't the AI just making things up?

Answers from notes must cite pages the AI was actually given. Invented citations are removed. If the notes don't cover something, it says so. Deadlines, plans and priorities are computed by ordinary code, not by the AI.

How is this different from Teams or Moodle?

Those hold the assignments. UniTask combines them, decides what to do first, plans the time, and teaches the content. It never submits on the student's behalf.

Can it read my Teams without permission?

No. The student signs in on Microsoft's own page and grants read-only access. If the college requires admin approval, UniTask shows that clearly instead of working around it.

Does it predict exam questions?

No. It shows how often topics appeared in the student's own past papers. That's history used for prioritising, and the app labels it that way.

What does the tutor remember, and can I delete it?

Short notes such as “confused DFA and NFA acceptance”. They are visible under **What it knows about me** and can be deleted one by one or all at once.

Is my data private?

Each student only sees their own data. Every API call checks ownership, files are in a private S3 bucket with short-lived links, and stored Teams tokens are encrypted.

What if the AI or an integration is down?

The app keeps working: the planner, deadlines, search and revision don't need the AI, and the tutor falls back to showing the relevant pages.

What does it cost to run?

Demo scale fits within AWS credits. The main variable cost is AI tokens, so AI calls are bounded and the cheaper model handles extraction.